

Yiming Yang, Ph.D.

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RESEARCH & PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate

Jun 2026 – Present

Oklahoma State University — Stillwater, OK, USA

School of Civil & Environmental Engineering

- **Rural Autonomous Vehicle (RAV) Development:** Evaluated zero-shot Vision-Language Models (VLMs), designed the full AV sensor suite, and architected the system pipeline.
- **Railway Safety & Sensing:** Designed a multimodal sensor fusion system (LiDAR, RGB, Thermal, 4D-Radar) for real-time railway trespassing detection and automated hazard alert generation.
- **Infrastructure Preparedness:** Conducted AV readiness assessments and physical/digital infrastructure evaluation to inform regional autonomous driving policy and deployment frameworks.
- **Grant Writing & Leadership:** Co-authored research proposals submitted to federal agencies (USDOT, FHWA) and mentored graduate students on sensor integration and software architecture.

EDUCATION

Ph.D. in Electrical and Computer Engineering

Aug 2021 – Apr 2026

Michigan Technological University — Houghton, MI, USA

- **Dissertation:** Robust Autonomous Driving in Winter Weather: Perception and Integrated Control Under Adverse Conditions

M.S. in Control Science and Engineering

Aug 2017 – Jul 2020

Chang'an University — Xi'an, China

- **Thesis:** Constrained Optimization and Distributed Model Predictive Control-Based Merging Strategies for Adjacent Connected Autonomous Vehicle Platoons

B.S. in Automation

Aug 2013 – Jul 2017

Chang'an University — Xi'an, China

- **Honors:** Selected for the National "Outstanding Engineers Education and Training Program"

RESEARCH INTERESTS

Resilient Perception & Embodied AI: Multimodal Sensor Fusion (LiDAR, Thermal, RGB), Adverse Weather Perception, Vision-Language Models (VLMs/VLAMs), Edge-Case Autonomy

Planning & Safety-Critical Control: Stochastic & Data-Driven Optimal Control, Model Predictive Path Integral Control (MPPI), Model-Based Design (MBD)

Connected & Automated Vehicles (CAVs): V2X Cooperative Platoon Control, Intelligent Transportation Systems (ITS), Traffic Simulation

PUBLICATIONS

Refereed Conference Proceedings

[C1] **Yiming Yang***, Jeremy P. Bos. "Beyond the Snowflakes: Disentangling Weather Effects from Confounding Meta-Factors in LiDAR Object Detection." In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026 (Accepted). [Link]

[C2] **Yiming Yang***, Jeremy P. Bos. "Tracking the Invisible: Self-Supervised Wheel Track Detection with Thermal-RGB Fusion." In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026 (Accepted). [Link]

[C3] **Yiming Yang***, Jeremy P. Bos. "Geometric and Radiometric Enhancements of Vehicles in LiDAR Point Clouds with Snow Accumulation." In *IEEE International Workshop on Metrology for Automotive (IEEE MetroAutomotive)*, Aug. 2026. [Link]

[C4] Akriti Yogi, Joshua Q. Li, **Yiming Yang**, Xinyang Du, Umesh Sharma, Kaustav Chatterjee, Amir Golalipour. "Transformer-Based Ride-Relevant Longitudinal Profile Reconstruction under Stop-and-Go Traffic Using Vehicle IMU Measurements." In *Transportation Research Board Annual Meeting (TRB)*, 2027 (Accepted).

[C5] Yiyang Zhou, Joshua Q. Li, **Yiming Yang**, Junlong Liu, Gary Yang, Amir Golalipour. "ResNet-Transformer Network for Continuous Pavement Friction Assessment Using Macro-Texture Profiles." In *Transportation Research Board Annual Meeting (TRB)*, 2027 (Accepted).

[C6] Junlong Liu, Xinyang Du, Haoran Yue, Yiyang Zhou, **Yiming Yang**, Joshua Q. Li, Amir Golalipour. "A Temporal Semi-Supervised Learning Framework for Drivable-Area Perception on Rural Roads." In *Transportation Research Board Annual Meeting (TRB)*, 2027 (Accepted).

Refereed Journal Articles

[J1] Joshua XS Li, Benjamin XY Li, **Yiming Yang***, Junlong Liu, Joshua Qiang Li, Anirudh Paranjothi. "Infrastructure Preparedness for Autonomous Vehicles in Rural Areas." *Sensors*, 26(19), 6067, 2026. DOI: 10.3390/s26196067. [Link]

[J2] **Yiming Yang***, Jeremy P. Bos. "Aggressive Autonomous Control on Snow and Ice." *SAE International Journal of Advances and Current Practices in Mobility*, Apr. 2025. DOI: 10.4271/2025-01-8040. [Link]

[J3] Haigen Min*, **Yiming Yang***, Yukun Fang, Pengpeng Sun, Xiangmo Zhao. "Constrained optimization and distributed model predictive control-based merging strategies for adjacent connected autonomous vehicle platoons." *IEEE Access*, vol. 7, Nov. 2019. DOI: 10.1109/ACCESS.2019.2952049. [Link]

Preprints & Manuscripts Under Review

[U1] **Yiming Yang**, Joshua Q. Li. "Evaluating Zero-Shot Vision-Language Models for Rural Autonomous Driving: Benchmarking, Robustness, and Prompt Engineering." Submitted to *IEEE Transactions on Intelligent Transportation Systems (ITS)*.

[U2] Junlong Liu, Kundan Parajulee, Xinyang Du, Yiyang Zhou, **Yiming Yang**, Joshua Q. Li, Amir Golalipour. "Deep Learning in Automated Pavement Surface Distress Assessment: A Systematic Survey." Submitted to *ASCE Journal of Computing in Civil Engineering*.

[U3] Chaz Cornwall, Logan Schexnaydre, Ian Mattson, **Yiming Yang**, Dillon Babcock, Nick Petersen, Jeremy P. Bos, Darrell Robinette. "Caerus Polaris: A Minimally-viable Autonomous Vehicle." Submitted to *SAE World Congress Experience (WCX 2027)*.

Open-Source Datasets & Software Tools

ThermalTrack: Multi-Modal Traversability Dataset [Dataset]

A dataset of 7,608 synchronized Thermal/RGB frames for resilient perception in whiteout and low-visibility winter conditions.

WADS-3D: LiDAR-Based Object Detection under Heavy Snow [Dataset]

Features 2,050 labeled LiDAR scans and 30,104 3D bounding boxes for object detection in heavy snowfall.

labelCloud-Enhanced: 3D Point Cloud Annotation Tool [GitHub]

High-efficiency 3D bounding box labeling with AI pre-labeling, spatial copy-paste, and enhanced viewport rotation.

RESEARCH EXPERIENCE

Grants & Research Proposals

Lead Author & Key Personnel, "Pavement Friction Assessment Using Automated and Connected Vehicle Data," *Federal Highway Administration (FHWA BAA)*, \$300,000, 2026. [Under Review]

Lead Author & Key Personnel, "Robust 3D Human Pose Estimation in Adverse Winter Conditions," *Sony Research Award Program*, \$100,000, 2026. [Under Review]

Research Experience

Postdoctoral Research Associate

June 2026 – Present

Oklahoma State University — Stillwater, OK

TRAVELS: Tribal & Rural Autonomous Vehicles for Efficiency, Livability and Safety

Sponsor: U.S. Department of Transportation (USDOT)

- Investigated AV policy constraints, legal barriers, and recommended legislative frameworks for autonomous driving.
- Evaluate vision-language models (VLMs) for edge-case perception and prompt engineering in rural area.
- Engineer and deploy autonomous vehicle platform, lead real-world field demonstrations across rural and tribal communities.

Graduate Research Assistant (Ph.D.)

Aug 2021 – Apr 2026

Michigan Technological University — Houghton, MI

SAE AutoDrive Challenge II (Sponsor: General Motors & SAE International)

- Architected the vehicle control interface and ROS-dSPACE communication pipeline via MicroAutoBox III.
- Developed safety-critical autonomy state machines with robust failure handling and safe-exit mechanisms.
- Led HIL/VIL validation and authored DVP&R documentation to track system-wide reliability.

Railroad Crossing Vehicle Warning (RCVW) System (Sponsor: Federal Railroad Administration - USDOT)

- Led V2X implementation, including VBS/RBS/RSU configuration, crossing map creation, and field testing.
- Developed an HIL system to accurately replicate and validate real-world connected vehicle scenarios.
- Presented large-scale PTV VISSIM simulations of the RCVW application for RLS and FRA web platforms.

UAS Data Integration for Transportation Infrastructure (Sponsor: Michigan Dept. of Transportation - MDOT)

- Curated datasets from UAV imagery (I-96 corridor) to train deep learning traffic counting models.
- Derived Origin-Destination (OD) metrics and verified system accuracy against state benchmarks.

Automated Infrastructure Generation (OSM to VISUM/VISSIM) (Dept. of Civil, Env., & Geospatial Eng.)

- Developed automated pipelines converting OpenStreetMap data into simulation-ready networks for PTV VISUM/VISSIM.
- Automated network connectivity, zone division, and connector generation, reducing modeling time from days to minutes.
- Conducted multi-scale OD analysis using Waymo (micro-level) and Wejo (macro-level) datasets.

Graduate Research Assistant (M.S.)

Aug 2017 – Jul 2020

Chang'an University — Xi'an, China

Autonomous Vehicle Control & Trajectory Tracking (Sponsor: National Key R&D Program / Shaanxi Province)

- Designed drive-by-wire algorithms for steering, propulsion, and braking using incremental PID.
- Deployed power supply, embedded systems for trajectory tracking algorithms, and remote-control hardware.
- Managed the gasoline-to-EV architecture transition and developed an MCU-based vehicle controller.

QLTD Intelligent Connected Expressway Testbase (Sponsor: Qilu Transportation Development Group)

- Implemented V2X-based adaptive Intelligent Driver Model (IDM) and Forward Collision Warning systems using DSRC/C-V2X.
- Validated V2V functionalities through real-world highway platoon tests and remote-control system deployment.

TEACHING EXPERIENCE

Co-Director / Co-Instructor

Aug 2026 – Dec 2026

Introduction to Autonomous Driving — Oklahoma State University

- Co-developed course curriculum and delivered lectures covering ROS/ROS2 middleware and the Autoware autonomous driving stack.
- Guided hands-on lab instruction spanning 1:10 scale autonomous vehicle development to full-scale autonomous vehicle platform integration.

Graduate Teaching Assistant

Jan 2023 – Apr 2026

Robotic Systems Enterprise — Michigan Technological University

- Mentored student sub-teams (*Controls & MathWorks*, *Safety & Testing*), running technical workshops and translating high-level objectives into actionable project sprints.
- **Controls Focus:** Guided deployment of steering, propulsion, braking, and trajectory-tracking algorithms using Model-Based Design (MBD) and Stateflow state machines.
- **Safety Focus:** Directed Safety of the Intended Functionality (SOTIF) validation and decoupled lateral/longitudinal controller testing for autonomous driving functions.

TALKS & PRESENTATIONS

[T1] "Tracking the Invisible: Self-Supervised Wheel Track Detection with Thermal-RGB Fusion." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Pittsburgh, PA, USA, Sep. 2026.

[T2] "Beyond the Snowflakes: Disentangling Weather Effects from Confounding Meta-Factors in LiDAR Object Detection." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Pittsburgh, PA, USA, Sep. 2026.

[T3] "Tracking the Invisible: Self-Supervised Wheel Track Detection with Thermal-RGB Fusion." *Oklahoma Transportation Research Summer Symposium (OTRSS)*, OKC, OK, USA, Jul. 2026.

[T4] "Beyond the Snowflakes: Disentangling Weather Effects from Confounding Meta-Factors in LiDAR Object Detection." *Robotics Colloquium, Michigan Tech Robotics Initiative*, Houghton, MI, USA, Feb. 2026.

[T5] "Aggressive Autonomous Control on Snow and Ice." *SAE World Congress Experience (WCX)*, Detroit, MI, USA, Apr. 2025. [Link]

[T6] "Deep Deterministic Policy Gradient based Cooperative Platoon Longitudinal Control Strategy." *Transportation Research Board Annual Meeting (TRB)*, Virtual, Jan. 2021. [Link]

[T7] "Research on Automatic Emergency Lane Change Control Strategy Based on MPC." *World Transport Convention (WTC)*, Beijing, China, Jun. 2019. [Link]

SELECTED HONORS & AWARDS

2nd Place , Concept Design Report & SRS: SAE/GM AutoDrive Challenge II (Year 4)	Jun 2025
2nd Place , Highway Challenge — SAE/GM AutoDrive Challenge II (Year 2)	Jun 2023
3rd Place , Overall Dynamics — SAE/GM AutoDrive Challenge II (Year 2)	Jun 2023
Outstanding Graduate , Chang'an University	Jun 2020
First-Class Academic Scholarship (2017–2018) , Chang'an University	Nov 2018
Outstanding Graduate Student Award (2017–2018) , Chang'an University	Nov 2018
Best Technology Award , i-VISTA Autonomous Driving Challenge	Aug 2018
First Prize , Urban Traffic Scene Challenge — i-VISTA Auto Driving Challenge	Aug 2018
Second Prize , General Service Robot Project — China Robot Competition	Aug 2018
First-Class Academic Scholarship (2016–2017) , Chang'an University	Nov 2017

SKILLS

Programming & Tools: Python, C++ , C, MATLAB, Linux/Bash, Git

Autonomy & Robotics: ROS/ROS 2, Autoware, Gazebo, V2X (DSRC/C-V2X), OpenStreetMap (OSM)

Perception & AI: PyTorch, OpenCV, Deep Learning, Vision-Language Models (VLMs), Sensor Fusion

Control & Simulation: Model-Based Design (MBD), Simulink/Stateflow, dSPACE (ControlDesk, ConfigurationDesk), PTV Vissim/Visum

Embedded Systems & HIL: Hardware-in-the-Loop (HIL/VIL), MicroAutoBox III, STM32, Teensy, CAN Bus

Hardware Design: SOLIDWORKS (CAD), Altium Designer (PCB Design)